Akoizo Original Practice - Form 1 - Answer key

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A bottle pusher's widest opening is 12.0 cm. What is its Pusher Bonus?

A. -34.5

B. -23.0

C. +34.5

D. -18.0

Correct: A

2. (2 points) What is the maximum vehicle width?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

A. 30.0 cm

B. 50.0 cm

C. 35.0 cm

D. 80.0 cm

Correct: C

3. (2 points) When does Run Time end?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. When the MP first passes the Target Point

B. Exactly at the Target Time

C. When the vehicle has been stationary for 3 seconds (the 3 s not counted)

D. When the bottle crosses the line

Correct: C

4. (2 points) Why must the pusher travel beyond the Bottle Line by more than zero?

A. The MP must cross the line

B. To reduce Run Time

C. The bottle’s near side must cross the line, so the pusher must go one bottle diameter past it

D. The rules require 10 cm past

Correct: C

5. (2 points) What is the first tiebreaker?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team’s first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. Better Distance Score of the scored run

B. Lower Time Score

C. Narrower pusher

D. Lighter vehicle

Correct: A

6. (4 points) Why might a team choose a wider pusher than the narrowest possible?

answer: A slightly wider pusher catches the bottle reliably despite lateral error, avoiding the huge cost of a miss. (4 points).

7. (4 points) Give two causes of wheel slip and one way to reduce it.

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

answer: Hard acceleration and sudden braking, or dusty floors; reduce slip with gradual speed ramps and clean tires. (4 points).

8. (4 points) Rank these by typical size of effect: Time Score, Pusher Bonus, construction violation.

Unit: Scoring a run: distance, time, bottle, and pusher.
Worked-case data: Target Time 14 s. Pusher opening 16.0 cm. The vehicle stops 22.5 cm from the Target Point in 16.37 s; the bottle ends 12.1 cm beyond the Bottle Line. Find the Run Score.
Separate investigation observations:
Pusher: Widest opening 20.0 cm.
Run 1: MP 6.0 cm off, bottle 5.0 cm past the line, time error 3.0 s.
Run 2: MP 3.0 cm off, bottle 8.0 cm short of the line, time error 0.5 s.

answer: Construction violation (+300) is largest, then Pusher Bonus (often -30 to -45), then Time Score (usually a few points). (4 points).

9. (4 points) What two things must a team using a kit provide?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team’s vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

answer: At least two functional modifications and an impounded Kit Modification Form documenting them, plus its own bottle pusher. (4 points).

10. (4 points) What must the team not do once a run starts?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

answer: Use, view, or access its programming tools until the vehicle is retrieved. (4 points).

11. (6 points) Solve this case and explain your reasoning: Target Distance 7.00 m, Bottle Line Distance 2.00 m, vehicle length 50 cm, bottle 6.5 cm, desired margin 3 cm. How far does the MP travel forward and backward?

reasoning: The line is at 9.00 m; the pusher face must reach 9.00 + 0.065 + 0.03 = 9.095 m.
MP forward travel = 9.095 - 0.50 = 8.595 m.
Reverse = 8.595 - 7.00 = 1.595 m. (3 points).

conclusion: The vehicle drives about 8.60 m forward and 1.60 m backward; the program should compute this from the posted values. (3 points).

12. (6 points) Solve this case and explain your reasoning: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?

reasoning: A second run is needed; without it, the second run counts as failed but the first still scores.
Fix the wire quickly, re-enter the posted values if needed, and place the vehicle ready-to-run.
Signal ready before time expires; a ready vehicle may complete its run. (3 points).

conclusion: The best score of two runs counts, so a quick, careful second run can only help if it is clean. (3 points).

13. (2 points) Which pusher should the team use?

Supplied synthetic observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. Pusher B

B. Pusher A

C. No pusher

Correct: A

14. (2 points) Define "Circumference" in the context of Electric Vehicle.

answer: π x wheel diameter; distance per revolution. (2 points).

15. (2 points) Define “Vehicle Distance” in the context of Electric Vehicle.

answer: Point-to-point distance from the MP to the Target Point. (2 points).

16. (2 points) Define “Energy storage device” in the context of Electric Vehicle.

answer: A capacitor or spring that could exceed battery power; banned. (2 points).

17. (2 points) Define “PID controller” in the context of Electric Vehicle.

answer: Feedback using proportional, integral, and derivative terms. (2 points).

18. (2 points) Define “Vehicle length offset” in the context of Electric Vehicle.

answer: Distance from the pusher face back to the MP. (2 points).